

COMP47670 Assignment 1

Autumn 2024

Summary:

The objective of this assignment is to collect a dataset from a public Web API and then use Python to prepare, analyse, and derive insights from the collected data.

The assignment should be implemented as two Jupyter Notebooks (not script files). Your notebooks should be clearly documented, using comments and Markdown cells to explain the code and interpret the results of your analysis.

Tasks:

Complete the following three tasks in two separate notebooks:

1. Data Collection

- Choose one of the APIs from the list of web APIs provided at this [link](#).
- In your first notebook, collect data from your chosen web API using Python. Depending on the choice of API, you might need to repeat the collection process multiple times to download sufficient data for analysis.
- Save the collected dataset in an appropriate format for subsequent analysis.

2. Data Preparation and Analysis

- In your second notebook, load the saved dataset from Task 1 into an appropriate data structure.
- Apply any data preprocessing steps that might be required to clean, filter or transform the dataset before analysis.
- Analyse, characterise, and summarise the cleaned dataset, using tables and visualisations where appropriate.
- Clearly explain each step of this process and interpret the results which are produced. Markdown cells should be used for this purpose.

3. Discussion

- At the end of your second notebook, discuss the following aspects of your assignment in Markdown cells (3-4 paragraphs):
 - Discuss any challenges faced in collecting data from the Web API.
 - Summarise the key insights gained from your analysis of the data.
 - Suggest ideas for further analysis which could be performed on the data (e.g. alternative analyses, integration of other sources of data).

Guidelines:

- The assignment should be completed individually. All submissions will be subject to plagiarism checking. Any evidence of plagiarism will result in a 0 grade.
- The grade awarded will depend on the complexity of the analysis and level of detail, i.e. data collection and preparation, analysis, interpretation etc.
- Submit your assignment via Brightspace. Your submission should be in the form of a single ZIP file which contains:
 1. Your two Jupyter notebooks (IPYNB files).
 2. The dataset you saved in Task 1. If your dataset is too large to upload to Brightspace, please include a smaller sample of the data in the ZIP file.
- While the assignment will be graded anonymously you should state your student number and the name of your chosen web API at the beginning of the notebooks.
- Penalties will apply for late submissions:
 - 1-5 days late: 1 grade point deduction, e.g. B to B-
 - 6-10 days late: 2 grade point deduction, e.g. B to C+
 - Assignments will not be accepted later than 10 calendar days without Extenuating Circumstances formally approved by UCD.